Politecnico di Torino

Master's Degree in Mathematical Engineering

Material for Thesis

Bibliography tracking

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Bibliography tracking

The following table provides an overview of the references used in the thesis, indicating whether each work has been read and whether it is currently cited in the written material.

Table 1: Tracking of GEO datasets, manifests, and probe-filtering resources

First author / Source	Dataset / Resource title	Read	Used	Ref.
GEO datasets				
NCBI	GSE69914: DNA methylation profiles in breast tissue samples	**Yes**	**Yes**	[1]
NCBI	GSE225845: High neighborhood deprivation impacts DNA methylation and gene expression in cancer-related genes	**Yes**	**Yes**	[2]
NCBI	GSE287331: DNA methylation patterns in breast cancer, benign tissue, and healthy controls	**Yes**	**Yes**	[3]
Manifests & annotation resources				
Illumina	Infinium MethylationEPIC v1.0 B5 Manifest File	**Yes**	**Yes**	[4]
Robinson	IlluminaHumanMethylation450kanno.ilmn12.hg19: annotation package for 450k arrays (Bioconductor)	**Yes**	**Yes**	[5]
Probe filtering & array-specific issues				
Naeem	Reducing the risk of false discovery enabling identification of biologically significant genome-wide methylation status using the HM450 array	**Yes**	**Yes**	[6]
Chen	Discovery of cross-reactive probes and polymorphic CpGs in the Illumina HumanMethylation450 microarray	**Yes**	**Yes**	[7]
Pidsley	Critical evaluation of the Illumina MethylationEPIC BeadChip microarray for whole-genome DNA methylation profiling	**Yes**	**Yes**	[8]
Zhou	Comprehensive characterization, annotation and innovative use of Infinium DNA methylation BeadChip probes	**Yes**	**Yes**	[9]
McCartney	Identification of polymorphic and off-target probe binding sites on the Illumina Infinium MethylationEPIC BeadChip	**Yes**	**Yes**	[10]

Table 2: Tracking of consulted and cited references for Medical / biological background

First author	Full paper title	Read	Cited	Ref.
Medical / biological background				
NIH	Understanding Cancer	**Yes**	**No**	[11]
NCI	BRCA Gene Changes: Cancer Risk and Genetic Testing Fact Sheet	**Yes**	**Yes**	[12]
Azizi	Introduction to Cancer Biology	**Yes**	**No**	[13]
Menon	Breast Cancer (StatPearls)	**Yes**	**No**	[14]
Hanahan	The Hallmarks of Cancer	**Yes**	**No**	[15]
Ehrlich	DNA Methylation in Cancer: Too Much, but Also Too Little	**Yes**	**Yes**	[16]
Bird	DNA Methylation Patterns and Epigenetic Memory	**Yes**	**Yes**	[17]
Russo	Epigenetic Mechanisms of Gene Regulation	**Yes**	**Yes**	[18]
Pina–Sanchez	Cancer Biology, Epidemiology, and Treatment in the 21st Century	**Yes**	**No**	[19]
Ji	Comprehensive Methylome Map of Lineage Commitment from Hematopoietic Progenitors	**Yes**	**No**	[20]
Wilcox	The CpG Landscape of Protein Coding DNA in Vertebrates	**Yes**	**No**	[21]

Table 3: Tracking of consulted and cited references for Data visualization and exploration

First author	Full paper title	Read	Cited	Ref.
Data visualization and exploration				
Maksimovic	SWAN: Subset-quantile Within Array Normalization for Illumina HumanMethylation450 BeadChips	**Yes**	**Yes**	[22]
Benelli	Charting Differentially Methylated Regions in Cancer with Rocker-meth	**Yes**	**No**	[23]
Seeboth	DNA Methylation Outlier Burden, Health, and Ageing in Generation Scotland and the Lothian Birth Cohorts	**Yes**	**No**	[24]
Zhao	A Urine-based DNA Methylation Assay, ProCUrE, to Identify Clinically Significant Prostate Cancer	**Yes**	**No**	[25]
Chen	Detecting Early-Warning Signals for Sudden Deterioration of Complex Diseases by Dynamical Network Biomarkers	**No**	**Yes**	[26]
West	On Dynamic Network Entropy in Cancer	**No**	**Yes**	[27]
Liu	Identifying the Critical States and Dynamic Network Biomarkers of Cancers Based on Network Entropy	**No**	**Yes**	[28]
Islam	Integration of DNA Methylation Patterns and Genetic Variation in Human Pediatric Tissues Helps Inform EWAS Design and Interpretation	**No**	**Yes**	[29]
Mansell	Guidance for DNA Methylation Studies: Statistical Insights from the Illumina EPIC Array	**No**	**Yes**	[30]
Chen	mLiftOver: Mapping DNA Methylation Data Across Array Platforms and Genome Builds	**No**	**Yes**	[31]

Table 4: Tracking of consulted and cited references for GSE69914 Applications

First author	Full paper title	Read	Cited	Ref.
GSE69914 Applications				
Chen	Discovery of cross-reactive probes and polymorphic CpGs in the Illumina HumanMethylation450 microarray	**Yes**	**No**	[7]
Teschendorff	DNA methylation outliers in normal breast tissue identify field defects that are enriched in cancer	**Yes**	**Yes**	[32]
Gao	The integrative epigenomic-transcriptomic landscape of ER-positive breast cancer	**Yes**	**No**	[33]
Ding	Integrative analysis identifies potential DNA methylation biomarkers for pan-cancer diagnosis and prognosis	**Yes**	**No**	[34]
Gao	DNA methylation patterns in normal tissue correlate more strongly with breast cancer status than CNVs	**Yes**	**No**	[35]
Croes	Large-scale analysis of DFNA5 methylation reveals its potential as biomarker for breast cancer	**Yes**	**No**	[36]
Fan	Prediction of disease genes based on stage-specific gene regulatory networks in breast cancer	**Yes**	**No**	[37]
Gao	Genome-wide discovery of circulating cell-free DNA methylation signatures for TNBC diagnosis	**Yes**	**No**	[38]
Zhu	An improved epigenetic counter to track mitotic age in normal and precancerous tissues	**Yes**	**No**	[39]
Yang	Pan-cancer integrative analysis of epigenetic enzymes reveals universal epigenomic deregulation	**Yes**	**No**	[40]
Bartlett	Inference of tissue relative proportions of breast epithelial subtypes (luminal progenitor, basal, luminal mature)	**Yes**	**No**	[41]
Teschendorff	EPISCORE: Cell-type deconvolution of bulk DNA methylomes using single-cell RNA-seq data	**Yes**	**No**	[42]

Table 5: Tracking of consulted and cited references for Data Pre–processing & quality control

First author	Full paper title	Read	Cited	Ref.
Data Pre–processing & quality control				
Du	Comparison of Beta-value and M-value methods for quantifying methylation levels by microarray analysis	**Yes**	**Yes**	[43]
Cazaly	Comparison of pre-processing methodologies for Illumina 450k methylation array data in familial analyses	**Yes**	**No**	[44]

Table 6: Tracking of consulted and cited references for Data Preparation

First author	Full paper title	Read	Cited	Ref.
Data Preparation				
Teschendorff	A beta-mixture quantile normalization method for correcting probe design bias in Illumina Infinium 450k DNA methylation data	**No**	**Yes**	[45]
de Almeida	A multimodal conversational agent for DNA, RNA and protein tasks	**No**	**Yes**	[46]
Sokolov	Decoding depression: a comprehensive multi-cohort exploration of blood DNA methylation using machine learning and deep learning approaches	**No**	**Yes**	[47]

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